STUDENT ASSIGNMENT

Title: Student Academic Performance – Data Cleaning, Feature Engineering & Power BI Visualization

Objective

This assignment requires students to use SAS Base Programming to audit, clean, validate, and engineer features from a student academic dataset. The final transformed dataset must be prepared and exported for visualization in Power BI.

PHASE 1 – Data Audit and Quality Assessment

- Import the dataset into SAS using a permanent library.
- Examine the dataset structure including variable names, types, and formats.
- Inspect the first observations to understand the data layout.
- Check for missing values in both numeric and character variables.
- Identify blank categories in categorical variables.
- Validate value ranges (e.g., attendance percentage, CGPA scale, age, study hours, sleep hours).
- Document all findings in a structured data quality report.

PHASE 2 – Data Cleaning

- Correct invalid or unrealistic numeric values where applicable.
- Handle missing values using appropriate techniques (removal, imputation, or flagging).
- Standardize categorical variables (e.g., gender labels).
- Detect and assess potential outliers.
- Justify whether outliers should be retained, capped, or removed.
- Create a cleaned version of the dataset.

PHASE 3 – Feature Engineering

- Create a performance category based on CGPA levels (High, Medium, Low).
- Create attendance level categories (Good, Moderate, Poor).
- Develop at least one engineered metric that reflects academic balance.
- Develop at least one productivity-related metric.
- Document all newly created variables and explain their purpose.
- Validate engineered features using statistical summaries.

PHASE 4 – Statistical Validation

- Perform descriptive statistics on key variables.
- Conduct correlation analysis to evaluate relationships.
- Build a regression model to predict Final CGPA.
- Evaluate model performance and identify significant predictors.
- Assess whether engineered features improve predictive performance.

PHASE 5 – Power BI Preparation

- Export the cleaned and engineered dataset into CSV format.
- Ensure the dataset is structured for visualization.
- Use clear and consistent column naming conventions.
- Verify no unnecessary variables are included in the final export.

PHASE 6 – Power BI Dashboard Requirements

- Create KPI cards showing average CGPA, average attendance, average study hours, and total students.
- Create visualizations comparing CGPA by gender and major.
- Create a scatter plot showing study hours versus CGPA with trend analysis.
- Visualize the performance category distribution.
- Include interactive slicers (Gender, Major, Performance Category, Attendance Level).
- Provide written insights explaining key findings.

Critical Thinking Questions

- Was the raw dataset completely clean? Justify your answer.
- What cleaning decisions were made and why?
- Which variables most strongly influence CGPA?
- Does social time negatively affect academic performance?
- Which group of students appears academically at risk?